

THE EX-MEN

In the **Punjab BJP** drawing rooms, the hot debate isn't about farmers or freebies. It's about the exes. Congress turncoats, led by **Sunil Jakhar** and **Capt. Amarinder Singh**, are wondering aloud about rekindling ties with the BJP's old flame, the Akalis. But Akali turncoats like Parminder Brar and Manjinder Sirsa won't have any of it. The old BJP hands, meanwhile, are quietly enjoying the tamasha.



Taken to Task



With the home portfolio no longer his, the IPS fraternity have been spared Bihar CM **Nitish Kumar's** exhaustive review meetings. The man himself has now redirected his energies. On November 28, the CM swooped on to the main secretariat, marching through departments while lecturing the IAS babus on punctuality, discipline, deadlines and more. Last heard, the IAS babus were wondering how the IPS officers had managed to get themselves off the hook.

SWITCHING OVER



At 77, **Sengottaiyan** briefly became the man of the moment in Tamil Nadu politics. His defection to actor Vijay's Thamizhaga Vetri Kazhagam days after AIADMK general secretary Edappadi Palaniswami expelled him—for suggesting a truce with rebel party groups—assured that. A lifelong AIADMK loyalist, Sengottaiyan even carried a photo of late CM J. Jayalalithaa in his shirt pocket when he went to meet Vijay. Anyway, strike one in TVK's favour in its battle to establish itself in Tamil Nadu politics.

Kaushik Deka with Arkamoy Datta Majumdar, Anilesh S. Mahajan, Amitabh Srivastava and Kavitha Muralidharan

TECH WATCH

By **Ajay Sukumaran**

WATER PURIFICATION A NEW SOLAR STILL

A 'solar still' is among the oldest water purifying devices used to distil potable water from saline or brackish water. The method is simple—it uses the sun's heat to evaporate saline water and collects the purified water via condensation.

While electricity-powered reverse osmosis (RO) systems, which push impure water through membranes at high pressure, are now the popular way of filtration, there has been substantial interest in improving on low-cost, passive desalination systems such as the solar still over the past decade. Driving this is a pressing global concern: according to a recent WHO-UNICEF report, 1 in 4 people globally (some 2.1 billion) still lack access to safe drinking water.

Researchers at the Indian Institute of Science (IISc), Bengaluru, have designed a solar-powered passive interfacial thermal desalination system that produces desalinated water without the use of electricity. The new system replaces capillary action (by which water climbs or is wicked up) with a siphon. The IISc project, funded by the Dept of Science and Technology, solved two challenges that such systems face. First, the issue of salt residue that gets left behind on the evaporator plate was sorted by controlling the flow rate of saline water; and the siphon enabled the system to scale up without height being a limiting factor.



Siphon-based desalination system

"Water scarcity is real, especially in coastal areas," says associate professor Susmita Dash, whose paper was published in the Elsevier journal *Desalination*, this year. "The idea is to make a portable system which you can carry with you, or for people in places where electricity is intermittent."